



Statistical quantification of potential impacts of upper-ocean stratification on tropical cyclone-induced sea surface cooling in the western North Pacific

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Abstract

Using high-resolution reanalysis products and observed tropical cyclones (TCs) in the western North Pacific during July–October from 1994 to 2015, this study investigates the relative importance of TC intensity and upper-ocean stratification in controlling sea surface cooling within 24 h before and after TC passages, with a focus on vertical mixing, a dominant process for the cooling. To minimize the influence of differences in TC translation speeds, only TCs with typical translation speeds are retained. The upper-ocean stratification is quantified by density-based virtual potential energy change (Δ VPE) that incorporates the effects of both mixed layer depth (MLD) and the density gradient beneath it. A statistical classification method reveals that a 43% difference in Δ VPE, associated with a shallower MLD and a steeper density gradient, leads to an additional surface cooling of 0.27°C for stronger TCs. In addition, correlation and regression analyses, including both strong and weak TCs and simply assuming that cooling amplitude is explained solely by TC intensity and Δ VPE, indicate that each predictor explains approximately 77% and 23% of the variance in cooling, respectively. Furthermore, Δ VPE outperforms four other metrics in estimating the impact of upper-ocean stratification on TC-induced cooling. These results provide a basis for improving the representation of ocean feedbacks in real-time TC intensity forecasts.

Keywords Tropical cyclone · Sea surface cooling · Upper-ocean stratification · Western North Pacific · Statistical analysis

1 Introduction

Sea surface cooling induced by the passage of a tropical cyclone (TC) (Leipper 1967) reduces the supply of moisture (latent heat flux) from the upper ocean, inhibiting TC development (Bender et al. 1993; Schade and Emanuel 1999; Ginis 2002; Cione and Uhlhorn 2003; Shen and Ginis 2003). To further understand TC-ocean interactions and improve TC intensity forecasts, the role of the upper ocean through this negative feedback has long been discussed. TC-induced sea surface cooling is primarily driven by vertical

mixing associated with the vertical shear of horizontal currents (Price 1981). This mixing entrains colder subsurface water into the mixed layer, and accounts for approximately 80% of the total cooling amplitude induced by a TC passage (Price 1981; Vincent et al. 2012a). Although the magnitude of the surface cooling is largely influenced by TC intensity, upper-ocean stratification prior to a TC passage also plays an important role (Vincent et al. 2012a; Cui et al. 2023; Looney and Foltz 2025). More specifically, a shallower mixed layer depth (MLD) and a steeper density gradient beneath it favor stronger surface cooling under a given TC forcing, and the effect of the subsurface density gradient becomes more important for stronger TCs (Yan et al. 2017). However, it is difficult to clearly distinguish the respective effects of MLD and the density gradient on the vertical-mixing process, because both factors jointly determine upper-ocean stratification. Furthermore, oceanic density is determined by both temperature and salinity. Especially in

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regions strongly influenced by precipitation or river inflow, haline stratification can also be important in addition to thermal stratification (Balaguru et al. 2012; Neetu et al. 2012; Yan et al. 2017; Miyagi and Tozuka 2024).

In this context, Vincent et al. (2012b) proposed a metric based on changes in upper-ocean potential energy caused by TCs. This measure is superior in the sense that it integrates the effects of both MLD and the density gradient beneath it, including contributions from both thermal and haline stratification, prior to TC passages. Using the measure, they showed that the cooling amplitude increases almost linearly with TC intensity, but depends strongly on upper-ocean stratification for a given TC wind forcing.

However, their study employed idealized axisymmetric TC wind patterns superimposed on six-hourly surface wind speeds from an atmospheric reanalysis product. Surface wind speeds may vary substantially within a few hours at a single location, suggesting that the six-hourly data may underestimate real TC intensity and subsequent upper-ocean responses. From an oceanic perspective, TC-induced oceanic responses are modulated by mesoscale eddies (Lin et al. 2005, 2008; Jaimes and Shay 2010; Ma et al. 2017). To adequately represent mesoscale eddies in ocean models, a horizontal resolution on the order of 0.1° is generally required (Paiva et al. 1999; Smith et al. 2000; Chelton et al. 2011). Since numerical simulations may randomly generate TCs over warm or cold eddies (Vincent et al. 2012a), assimilation of observational data is also important. Although quantifying eddy effects themselves is beyond the scope of this study, a reanalysis product with a high horizontal resolution of about 0.1° is more suitable for representing mesoscale variability.

Similarly, most previous studies analyzed TC-induced surface cooling over several days before and after TC passages. This timescale reflects the fact that sea surface temperature (SST) variability associated with TC passages can be broadly divided into undisturbed, forcing, and relaxation stages (Mei and Pasquero 2013; Cui et al. 2023). Until a few days before a TC arrival, SST remains relatively stable. However, the SST gradually responds to TC forcing from approximately one day before TC arrival, and the cooling typically reaches its maximum amplitude approximately one day after the TC passage. Subsequently, the cooling diminishes due to the solar heating or horizontal advection. This recovery timescale ranges from one day to several weeks. Thus, the appropriate analysis window depends on which stage a study aims to investigate. Yet, due to the limited number of direct observations under TCs, many previous studies have relied on relatively low-temporal-resolution reanalysis products, when investigating SST responses to TCs.

Motivated by the above, this study aims to quantify the potential impacts of upper-ocean stratification on TC-induced sea surface cooling during the forcing stage (i.e., from one day before to one day after TC passages), when TC development is strongly affected by SST cooling beneath the TC (Emanuel 1999b; Cione and Uhlhorn 2003). For this purpose, we use hourly and three-hourly reanalysis products for surface winds and the upper ocean, respectively, together with approximately 7,000 points passed by observed TC centers over more than 20 years in the western North Pacific, the region with the highest TC frequency worldwide (Gray 1968).

Since observed surface cooling is typically stronger on the right (left) side of a TC track in the Northern (Southern) Hemisphere (Stramma et al. 1986; Price et al. 2008; D'Asaro et al. 2014), it is necessary to account for this asymmetric response while avoiding excessive sensitivity to variability at individual grid points. For this purpose, we calculate spatially-averaged parameters centered on individual grid points passed by TC centers. This framework enables a more statistically robust quantification of the relationship between upper-ocean stratification and TC-induced cooling.

This paper is organized as follows. The next section describes the observational and reanalysis datasets and their validations. The methods used to quantify TC intensity and the potential impacts of upper-ocean stratification, and their statistical classifications, are also explained. Section 3 presents the main results and supporting analyses. The final section provides a summary and discusses potential implications of this study.

2 Data and methods

All analyses are conducted within 10°N – 30°N , 120°E – 150°E , a region with high TC frequency in the western North Pacific, during the main TC season from July to October (Gao et al. 2022). The analysis period (1994–2015) is selected to ensure consistency among all datasets, as the ocean reanalysis product used in this study is available only during this period.

2.1 TC position and translation speed

Positions of observed TC centers are obtained from the best track data provided by the Regional Specialized Meteorological Center in Tokyo. Their latitude and longitude are given at 0.1 -degree intervals. The temporal resolution is ordinarily six-hourly, but is sometimes three-hourly when TCs rapidly intensify or make landfall. The maximum

sustained wind speed of tropical depressions and extratropical cyclones is defined as 0 knots, and the minimum positive maximum sustained wind speed is 35 knots. To focus on relatively large cooling events and to exclude extratropical cyclones, we have decided to use TCs with maximum sustained wind speeds of 35 knots or higher. For each TC, the translation speed (U) is calculated at each position by dividing the distance traveled over the preceding three or six hours by the corresponding time interval.

In our main analyses, each variable (i.e., surface wind energy, MLD, potential density gradient beneath the MLD, virtual potential energy change, and SST introduced in the following subsections) is spatially averaged within a $2^\circ \times 2^\circ$ box centered on each point passed by a TC center, but is not temporally averaged along the TC track. In validations of reanalysis products with buoy and satellite-based observations, however, we use variables at individual grid points passed by TC centers.

2.2 Surface wind

2.2.1 Data description

Hourly surface wind speeds at a height of 10 m are obtained from the National Centers for Environmental Prediction (NCEP) Climate Forecast System Reanalysis (CFSR) during 1994–2010 (Saha et al. 2010) and the NCEP Climate Forecast System Version 2 (CFSv2) during 2011–2015 (Saha et al. 2014). The CFSR was produced using a coupled model system consisting of the Global Forecast System (GFS) for the atmosphere, the Modular Ocean Model version 4p0d (MOM4) (Griffies et al. 2004) for the ocean, the Noah land-surface model (Ek et al. 2003), and a three-layer sea-ice model from the Sea Ice Simulator at the Geophysical Fluid Dynamics Laboratory. Each component has a tailored data assimilation system: the Grid-point Statistical Interpolation (GSI) for the atmosphere, the Global Ocean Data Assimilation System (GODAS) for the ocean, and the Global Land Data Assimilation System (GLDAS) for the land surface. For the atmosphere, in addition to observational data in the upper layers obtained from radiosondes, pibals, and aircrafts, satellite-based surface winds and temperatures were also used in the assimilation.

The CFSR data were used as initial states to produce retrospective forecasts in CFSv2. Although the ocean and sea-ice models in CFSv2 are identical to those in CFSR, the atmospheric and land components differ slightly owing to updates in physical parameterizations. On the other hand, initial conditions for real-time operational forecasts in CFSv2 were obtained from the real-time operational

Climate Data Assimilation System version 2, which is a real-time continuation of CFSR.

2.2.2 Processing

Horizontal resolutions of the wind speeds in CFSR and CFSv2 (hereafter, CFS as a collective name) are approximately $0.2^\circ \times 0.2^\circ$ and $0.3125^\circ \times 0.3125^\circ$, respectively. Since positions of TC centers are given at 0.1-degree intervals, we have linearly interpolated the original wind speeds to $0.1^\circ \times 0.1^\circ$.

2.2.3 Surface wind energy

To quantify the impact of TC intensity on TC-induced sea surface cooling (ΔSST), we calculate surface wind energy (E_w) defined as follows:

$$E_w = \int_{-24 \text{ hour}}^{+24 \text{ hour}} \rho_a C_d V^3 dt. \quad (1)$$

Here, ρ_a ($=1.26 \text{ kg/m}^3$) is the density of the air, C_d ($=2 \times 10^{-3}$) is the drag coefficient, V is the interpolated surface wind speed (m/s), and dt ($=1 \text{ h}=3,600 \text{ s}$) is the temporal interval. The drag coefficient is known to vary with wind speed and tend to level off at high wind speeds (Large and Pond 1981; Sullivan et al. 2012). However, its magnitude generally remains on the order of 10^{-3} . In contrast, the impact of surface wind speed itself is proportional to the cube of wind speed. Therefore, E_w , defined as the time integration of $\rho_a C_d V^3$, is primarily controlled by V rather than by relatively small variations in C_d . In addition, assuming a constant value of C_d simplifies the interpretation of the relationship between surface wind energy and SST cooling. For these reasons, we have assumed a constant C_d in this study.

This formulation of E_w is similar to the temporal integration of power dissipated by surface friction (PD) (Bister and Emanuel 1998; Emanuel 1999a):

$$PD = \int_{t_0}^{t_c} \rho_a C_d V^3 dt, \quad (2)$$

where t_0 is the time when a TC starts influencing a given location, and t_c is the current time. Since PD is recognized as a good proxy of kinetic energy transferred from TC winds to surface currents in the ocean (Vincent et al. 2012b), we apply this concept within 24 h before and after a TC passage: $t_0 = -24 \text{ h}$ and $t_c = +24 \text{ h}$. We note that E_w is positively correlated with the maximum sustained wind speed,

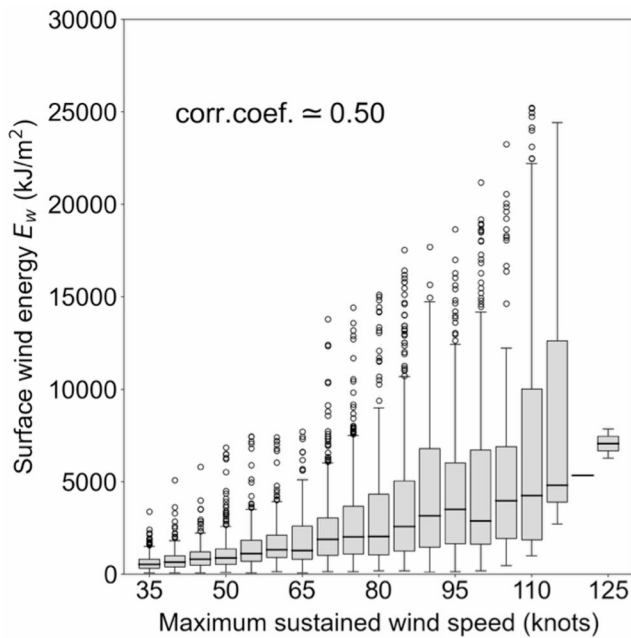


Fig. 1 Relationship between maximum sustained wind speed (knots) obtained from the Regional Specialized Meteorological Center in Tokyo and surface wind energy (E_w) (kJ/m^2) calculated using wind speeds of the National Centers for Environmental Prediction (NCEP) Climate Forecast System (CFS). Each grey box indicates the interquartile range (IQR), with a horizontal black line at the median. Each whisker extends up to 1.5 times the IQR, with outliers plotted as fliers. The correlation coefficient between maximum sustained wind speeds and E_w values is calculated using all values, including the quartiles, medians, and outliers

with a correlation coefficient of 0.50 (Fig. 1), supporting its use as a metric for TC intensity in this study.

2.2.4 Validation

To evaluate the reliability of wind speeds and surface wind energy associated with observed TCs, these values are compared with scalar wind speeds observed by the Kuroshio Extension Observatory (KEO) buoy (32.3°N, 144.6°E) (Cronin et al. 2008), measured at a height of 4 m, within one day before and after TC passages (Fig. 2). The KEO buoy is a reference station whose observations are not assimilated into the atmospheric and oceanic products in this study, and thus its data are appropriate for independent validation. Since the buoy was deployed in 2004, hourly data from July to October during 2004–2015 are used. Because this study focuses on wind variability near TC centers, eight TC cases passing within 100 km of the buoy are identified (Table 1). Surface wind data associated with the passage of T1514 are missing in the buoy record. Thus, the following comparisons exclude this case. As a result, CFS wind speeds and surface wind energy are largely consistent with the buoy observations within the 48-hour window: the correlation coefficients are 0.84 and 0.96, respectively, and the root mean squared errors (RMSEs) are 3.3 m/s and 325 kJ/m^2 , respectively. Furthermore, the time series of surface winds from 10 days before to 4 days after TC passages (Fig. 3)

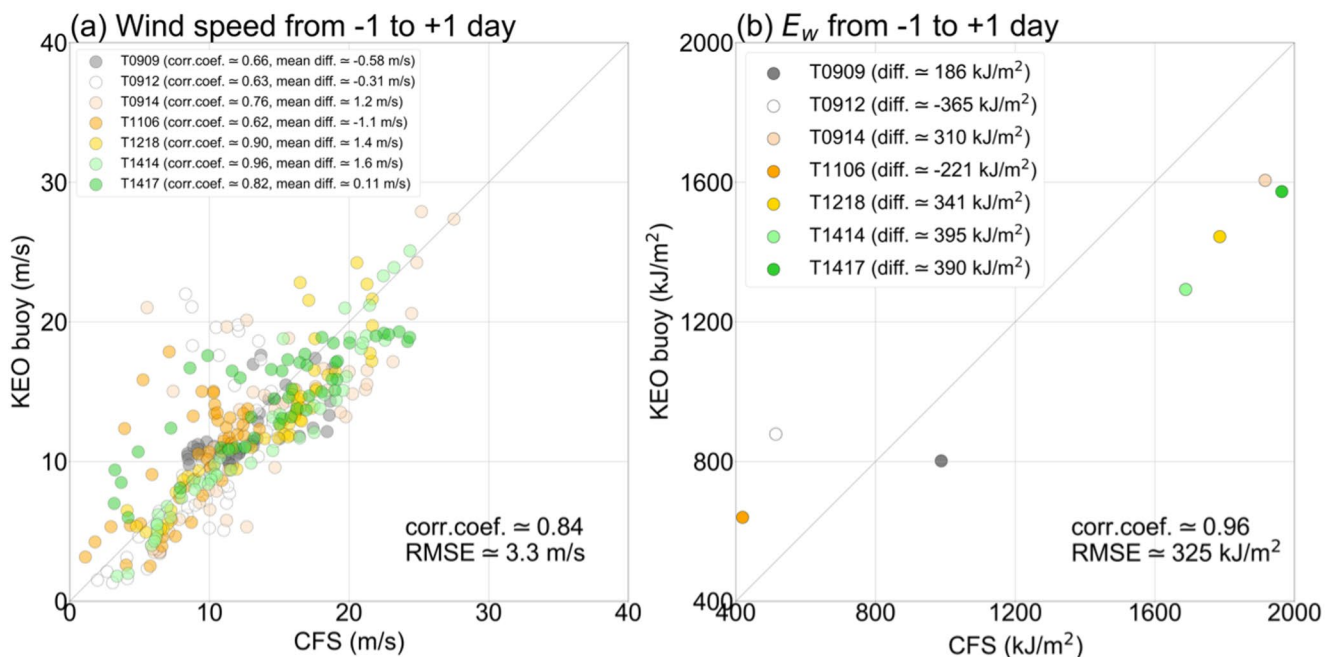


Fig. 2 Comparison of (a) wind speeds (m/s) and (b) E_w (kJ/m^2) from one day before to one day after the passage of seven tropical cyclones (TCs) between CFS and the Kuroshio Extension Observatory (KEO) buoy, from July to October during 2004–2015. For each TC case, indi-

cated by its international number ID, the differences in wind speeds and E_w (CFS minus KEO buoy) are shown in the upper left of the panels. Their correlation coefficient and root mean squared error (RMSE), calculated using all cases, are shown in the lower right of each panel

Table 1 Minimum distances between centers of tropical cyclones (TCs) and the Kuroshio Extension Observatory (KEO) buoy for eight TCs from July to October during 2004–2015

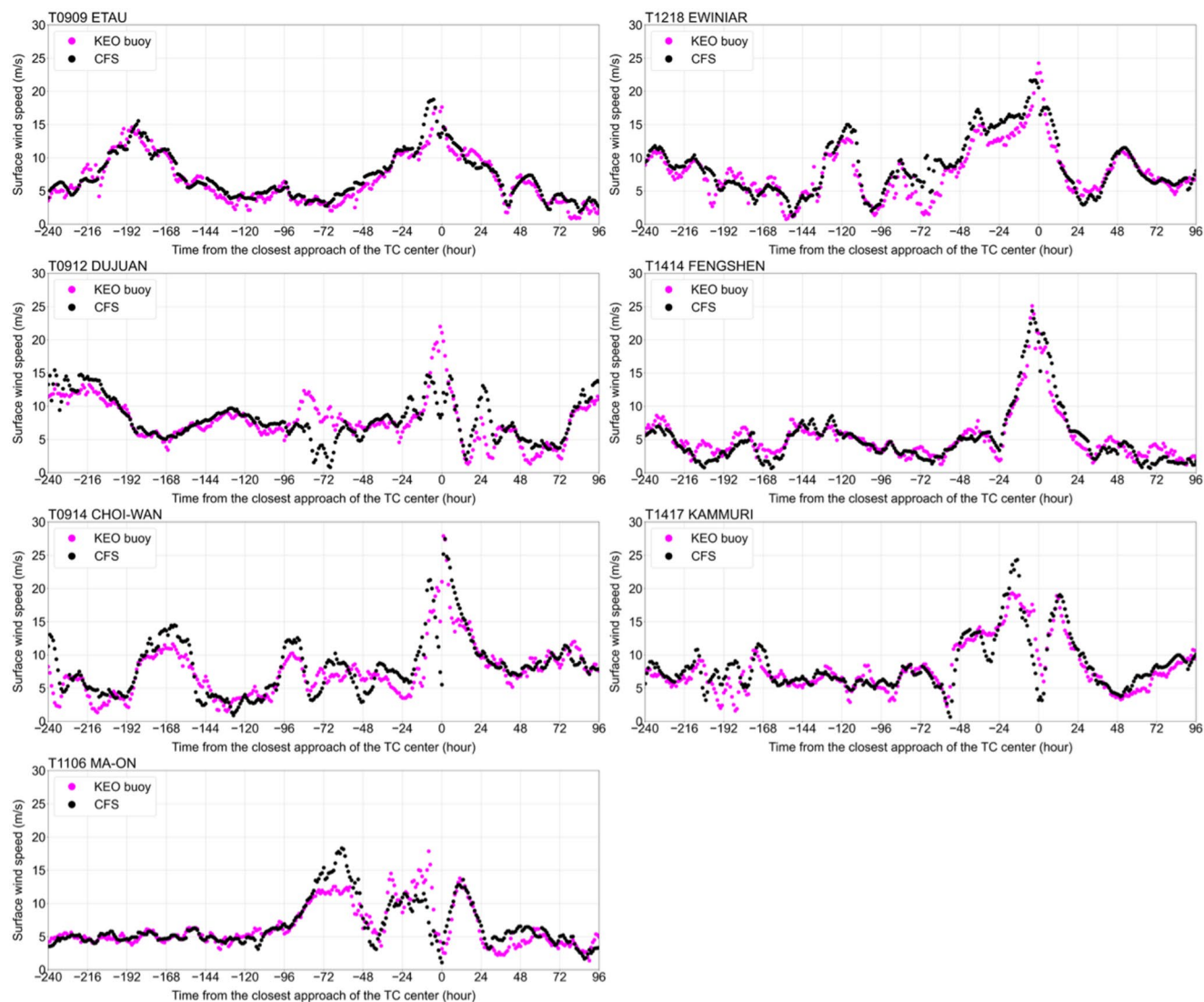
TC International number ID and name	Minimum distance to the KEO buoy
T0909 ETAU	72 km
T0912 DUJUAN	47 km
T0914 CHOI-WAN	65 km
T1106 MA-ON	39 km
T1218 EWINIAR	50 km
T1414 FENGSHEN	91 km
T1417 KAMMURI	78 km
T1514 MOLAVE	29 km

show that the temporal evolution of wind variability is mostly consistent between CFS and the buoy observations.

2.3 Temperature and salinity in the upper ocean

2.3.1 Data description

To investigate upper-ocean stratification before TC passages, we use three-hourly temperature and salinity from the Global Ocean Forecasting System (GOFS) 3.1 (Cummings 2005; Cummings and Smedstad 2013; Helber et al. 2013). This ocean reanalysis product is produced using the Hybrid Coordinate Ocean Model (HYCOM) and the Navy Coupled

**Fig. 3** Time series of surface wind speeds from 10 days before to 4 days after the passage of seven TCs, comparing CFS (black dots) and observations from the KEO buoy (magenta dots)

Ocean Data Assimilation (NCODA) system, with surface forcing from CFS. The NCODA system (Cummings 2005; Cummings and Smedstad 2013) applies a three-dimensional variational scheme using 24-hour model forecasts as the first guess. It then assimilates various observational data, including satellite altimetry, both satellite-based and in-situ SST observations, and vertical temperature and salinity profiles obtained from expendable bathythermographs (XBTs), Argo floats, and moored buoys. In addition, the Improved Synthetic Ocean Profiles method converts surface observations into subsurface profiles within the water column (Helber et al. 2013).

2.3.2 Processing

Although GOFS 3.1 has a horizontal resolution of 0.08° with 40 vertical levels, our analyses are confined to the upper 200 m to focus on upper-ocean responses to TCs. This depth range is sufficient, as MLD—defined by a potential density increase of 0.125 kg/m^3 from the surface—is consistently shallower than 200 m within 24 h before and after TC passages. The temperature and salinity profiles in the upper 200 m are available at non-uniform depths: 0, 2, 4, 6, 8, 10, 12, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100, 125, 150, and 200 m. To calculate the parameters used in this study, the profiles are first linearly interpolated to 1-m intervals.

2.3.3 Virtual potential energy change

To quantify the impact of upper-ocean stratification on ΔSST , we introduce the virtual potential energy change ΔVPE (Vincent et al. 2012b), defined by

$$\Delta \text{VPE} = \int_{-h_m}^{0 \text{ m}} (\rho_f(T_f, S_f) - \rho_i(T_i, S_i)) g z dz. \quad (3)$$

Here, potential density of sea water ρ is derived from potential temperature T and salinity S , the subscripts i and f represent 24 h before and after a TC passage, respectively, g is the acceleration due to gravity, z is depth, dz is the interval of depth ($=1 \text{ m}$), and h_m is the virtual MLD after the TC passage. All parameters are calculated at a three-hourly interval. The density profiles prior to TC centers' arrival (ρ_i) are calculated from the original temperature (T_i) and salinity (S_i) profiles, while the density profile after the TC passage (ρ_f) is estimated as the vertically averaged ρ_i values from the surface to h_m . Here, h_m is defined as the depth at which the vertically averaged potential temperature from the surface equals $\text{SST}|_{-24 \text{ h}} + \text{virtual } \Delta \text{SST} (T_v)$. Therefore, the calculation of ΔVPE requires only profiles before TC centers' arrival and can be applied without profiles after

the arrival. Here, we adopt $T_v = -1^\circ \text{C}$ to represent the typical cooling magnitude, because the mean SST cooling amplitude at individual points within 24 h before and after TC passages in this study is approximately -1°C , which is broadly consistent with previous studies (e.g., Vincent et al. 2012b; Cui et al. 2023).

2.3.4 Density gradient beneath MLD

We define the density gradient beneath MLD (α) as the difference in potential density per meter between the MLD and h_m . Since oceanic density typically increases with depth, the density gradient is generally negative.

2.3.5 TC-induced sea surface cooling

TC-induced sea surface cooling ΔSST is defined as

$$\Delta \text{SST} = \text{SST}|_{+24 \text{ hour}} - \text{SST}|_{-24 \text{ hour}}. \quad (4)$$

To focus on the mixed layer deepening through vertical mixing, which is a dominant process for ΔSST (e.g., Price 1981), our analyses are limited to cases with positive MLD change ($\text{MLD}|_{-24 \text{ h}} < \text{actual MLD}|_{+24 \text{ h}}$) and negative ΔSST , thereby excluding surface warming due to solar radiation or horizontal advection of water.

2.3.6 Validation

Similar to the validation for surface winds, we compare ΔSST and ΔMLD (actual $\text{MLD}|_{+24 \text{ h}}$ minus $\text{MLD}|_{-24 \text{ h}}$) induced by eight TCs, including an additional sample T1514, between GOFS and the buoy observations (Fig. 4). Regarding the buoy in the upper 200 m, the temperature profiles are available at 1, 5, 8, 10, 15, 18, 20, 25, 35, 38, 40, 50, 75, 100, 105, 125, 150, 175, and 200 m, and the salinity profiles are available at 1, 5, 10, 15, 20, 25, 35, 40, 50, 75, 100, 125, 150, 175, and 200 m. To compare the observations and GOFS in the same vertical interval, the variables of the buoy are first linearly interpolated to 1-m intervals. It is found that GOFS and the observations are largely consistent: the correlation coefficients for ΔSST and ΔMLD are 0.94 and 0.85, respectively, and the RMSEs are 0.24°C and 4.7 m, respectively. Furthermore, SST values from 10 days before to 4 days after the TC passages are also mostly consistent between the two datasets (not shown). These results indicate that GOFS reasonably captures the observed short-term variability in the upper ocean during TC passages.

However, the above validation is limited to a single point passed by a few TCs. To further evaluate SST obtained from GOFS over the whole region (10°N – 30°N , 120°E – 150°E), we also compare them with daily satellite-based observations

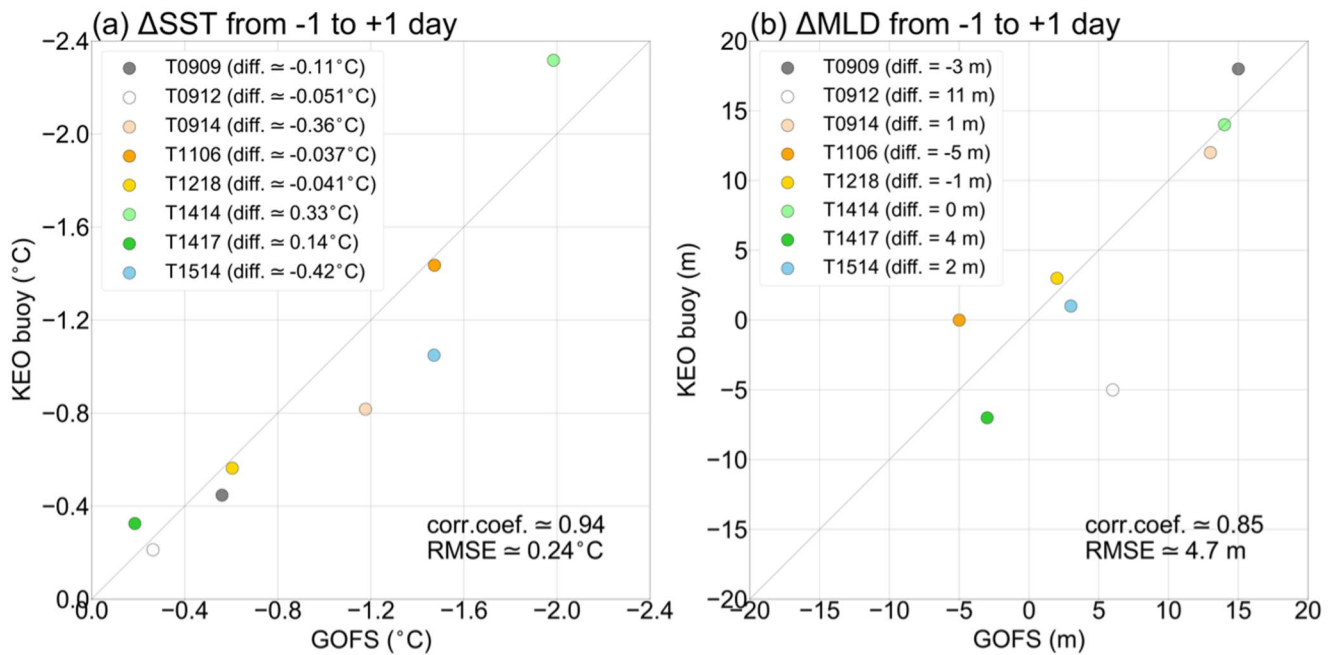


Fig. 4 Comparison of changes in (a) sea surface temperatures (Δ SST) ($^\circ\text{C}$) and (b) mixed layer depths (Δ MLD) (m) from one day before to one day after the passage of eight TCs, between the Global Ocean Forecasting System (GOFS) 3.1 and observations from the KEO buoy, from July to October during 2004–2015. For each TC case,

indicated by its international number ID, the differences in Δ SST and Δ MLD (GOFS minus KEO buoy) are shown in the upper left of the panels. Their correlation coefficient and RMSE calculated for all cases are shown in the lower right of each panel

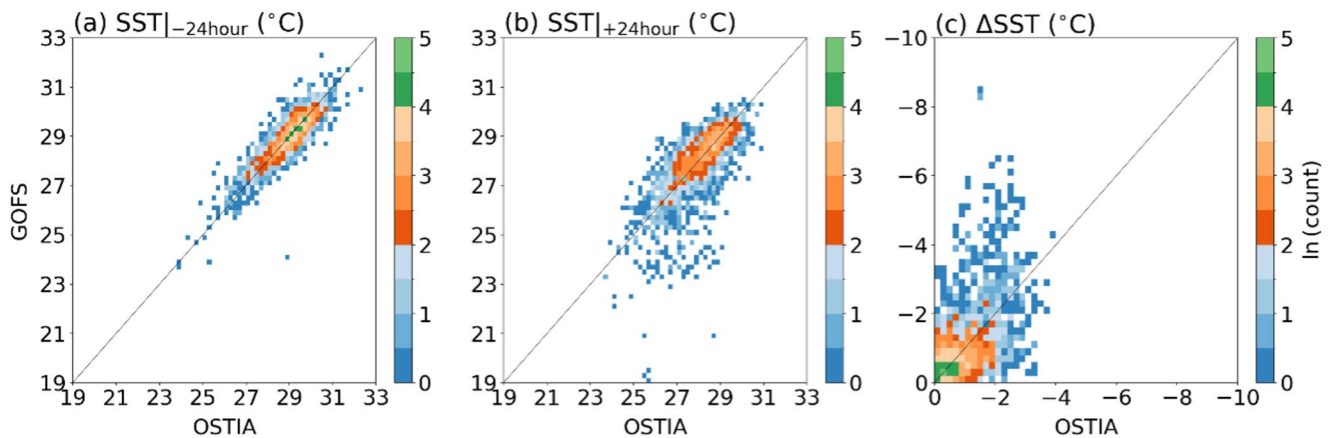


Fig. 5 Comparison of (a) $\text{SST}|_{-24\text{h}}$, (b) $\text{SST}|_{+24\text{h}}$, and (c) Δ SST between GOFS and the reprocessed Operational Sea Surface Temperature and Ice Analysis (OSTIA) system, from July to October during

1994–2015. The color indicates the natural logarithm of the number of samples within each $0.2^\circ\text{C} \times 0.2^\circ\text{C}$ bin

from the reprocessed Operational Sea Surface Temperature and Ice Analysis (OSTIA) system (Worsfold et al. 2024). The horizontal resolution of OSTIA is $0.05^\circ \times 0.05^\circ$, which is also linearly interpolated to $0.1^\circ \times 0.1^\circ$ in this study. The GOFS tends to overestimate TC-induced cooling (Fig. 5c), likely due to the known underestimation of cooling in OSTIA (Nakada and Tomita 2025); the correlation coefficient and RMSE between GOFS and OSTIA are 0.49 and 0.87°C , respectively. However, SSTs one day before and after TC passages are largely consistent (Figs. 5a and b):

their correlation coefficient and RMSE for $\text{SST}|_{-24\text{h}}$ are 0.87 and 0.47°C , respectively, and those for $\text{SST}|_{+24\text{h}}$ are 0.73 and 0.91°C .

In addition, it is important to confirm that the spatial distribution of upper-ocean stratification in GOFS is consistent with observations. Figures 6, 7 and 8 therefore compare the 22-year-mean MLD, α , and Δ VPE in each month between GOFS and the monthly objective analyses of the EN 4.2.2 c14 (Good et al. 2013; Cheng et al. 2014; Gouretski and Cheng 2020) (hereafter, EN4). Since the horizontal

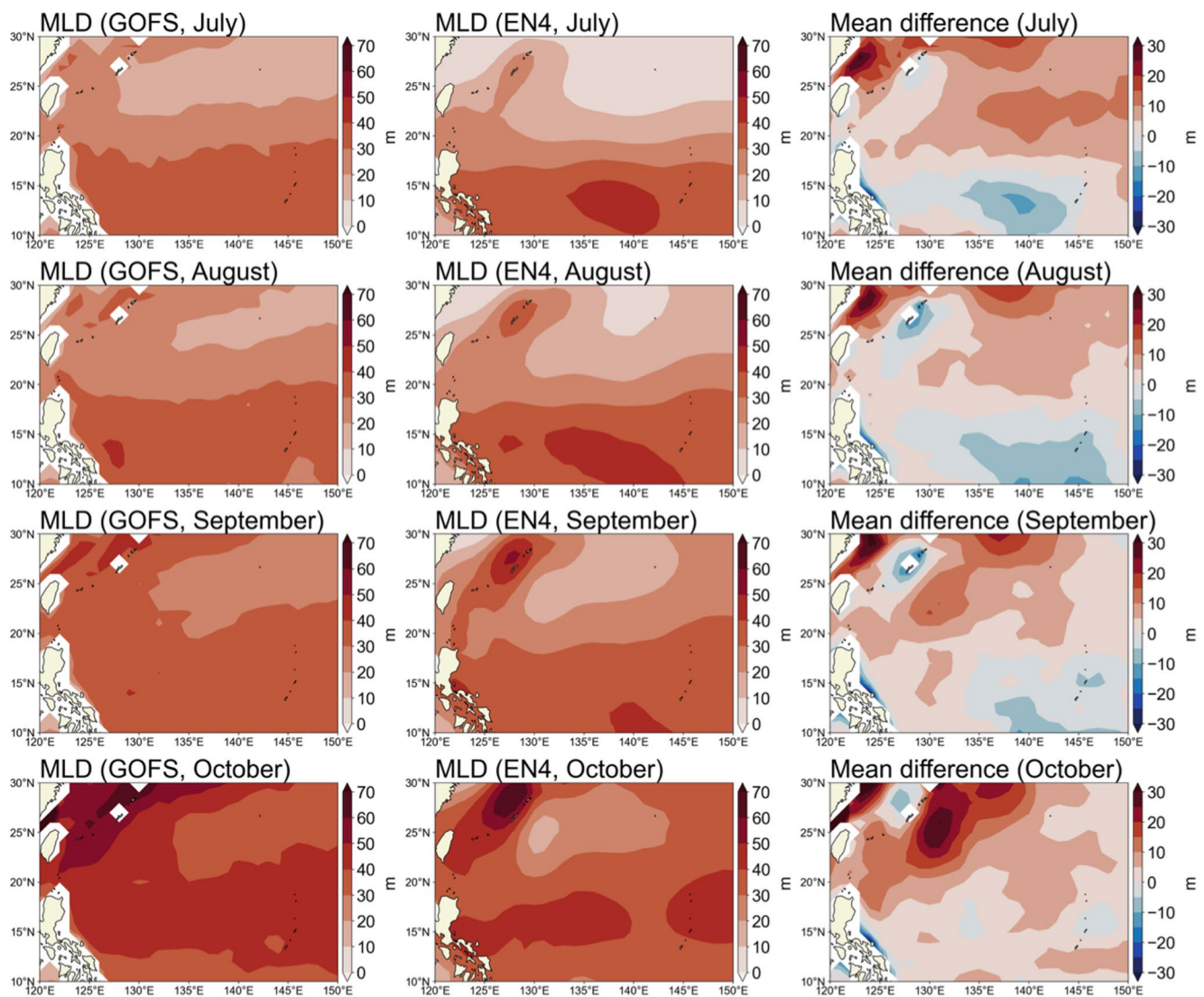


Fig. 6 Spatial distribution of the 22-year-mean MLD (m) obtained from GOFS (left panels) and the EN 4.2.2 c14 (middle panels) and their differences (GOFS minus EN4; right panels) for each month from July to October during 1994–2015

resolution of EN4 is $1^\circ \times 1^\circ$, variables of GOFS are linearly interpolated to $1^\circ \times 1^\circ$. The temperature and salinity profiles of EN4 from the surface to approximately 200 m are available at 5.02159, 15.07854, 25.16046, 35.27829, 45.44776, 55.69149, 66.04198, 76.54591, 87.27029, 98.31118, 109.8062, 121.9519, 135.0285, 149.4337, 165.7285, 184.6975, and 207.4254 m. To compare EN4 and GOFS in the same vertical interval, the variables of EN4 at the 17 depths are also linearly interpolated to 1-m intervals from 0 to 200 m. Although the magnitudes of each variable (MLD, α , and ΔVPE) differ somewhat at individual grid points, their qualitative spatial distributions are largely consistent.

2.4 Classification methods

To quantify the relative importance of surface wind energy and upper-ocean stratification on ΔSST , we have compared ΔSST among four subgroups: “strong TC and large ΔVPE ”, “strong TC and small ΔVPE ”, “weak TC and large ΔVPE ”, and “weak TC and small ΔVPE ”. The two strong-TC (weak-TC) subgroups are selected so that their E_w values are comparable, i.e., the difference between the two-subgroup means is not statistically significant, while their ΔVPE values are statistically different. Similarly, large (small) ΔVPE subgroups are selected so that their Δ

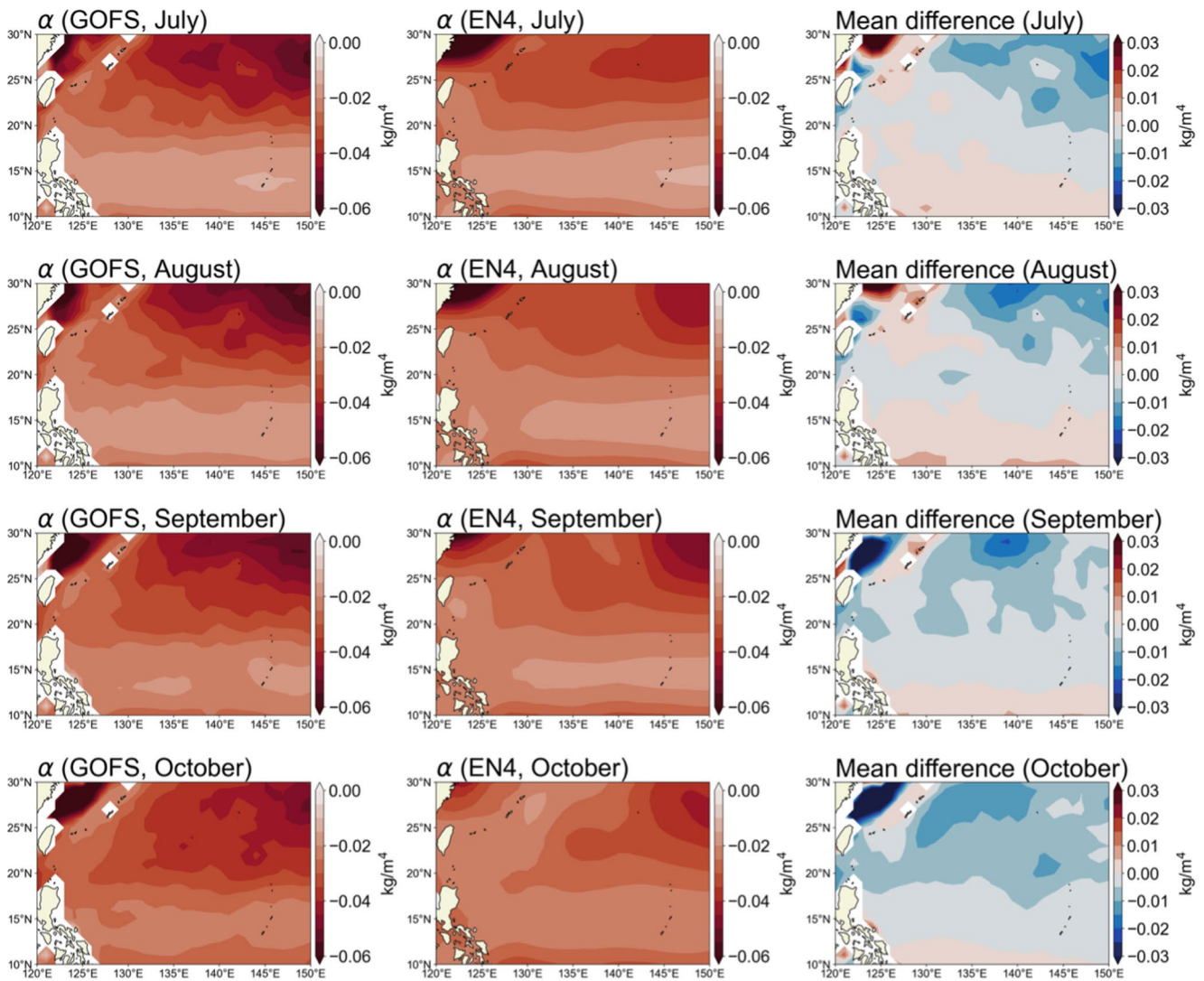


Fig. 7 As in Fig. 6, but for the potential density gradient beneath MLD (α) (kg/m^4)

VPE values are comparable, while their E_w values are statistically different. This classification enables evaluation of the effects of Δ VPE and E_w on Δ SST under comparable TC intensity or upper-ocean stratification.

The criteria used to define large and small E_w and Δ VPE are summarized in Table 2. We note that the thresholds of the 20th, 50th, and 80th percentiles were therefore chosen as a compromise between achieving sufficient separation among categories and maintaining adequate sample sizes. With these thresholds, each category (strong TC, weak TC, large Δ VPE, and small Δ VPE) contains more than 100 samples, allowing statistically meaningful comparisons. We also confirmed that the qualitative results are robust when slightly different percentile thresholds are used. To minimize the influence of differences in TC translation speed, we restrict the analysis to TCs moving at nearly the same

speed, within the range of U_{30} and U_{70} , where the subscript m denotes the m th percentile of U . Although TC translation speed is restricted to typical values and TC size is not explicitly considered, their effects may be partly reflected in surface wind energy within the common analysis window (i.e., from one day before to one day after TC passages) and $2^\circ \times 2^\circ$ analysis box. Because it is difficult to isolate the individual contributions of these effects, in a physically interpretable manner, the present study focuses on the relative importance of surface wind energy and upper-ocean stratification. Figure 9 shows the histograms of each parameter, along with their percentiles used for classification.

As shown in Fig. 8, the 22-year-mean Δ VPE values tend to decrease with latitude, and are large along the western boundaries, which are qualitatively consistent between GOFS and EN4. To the north (south) of 20°N, GOFS

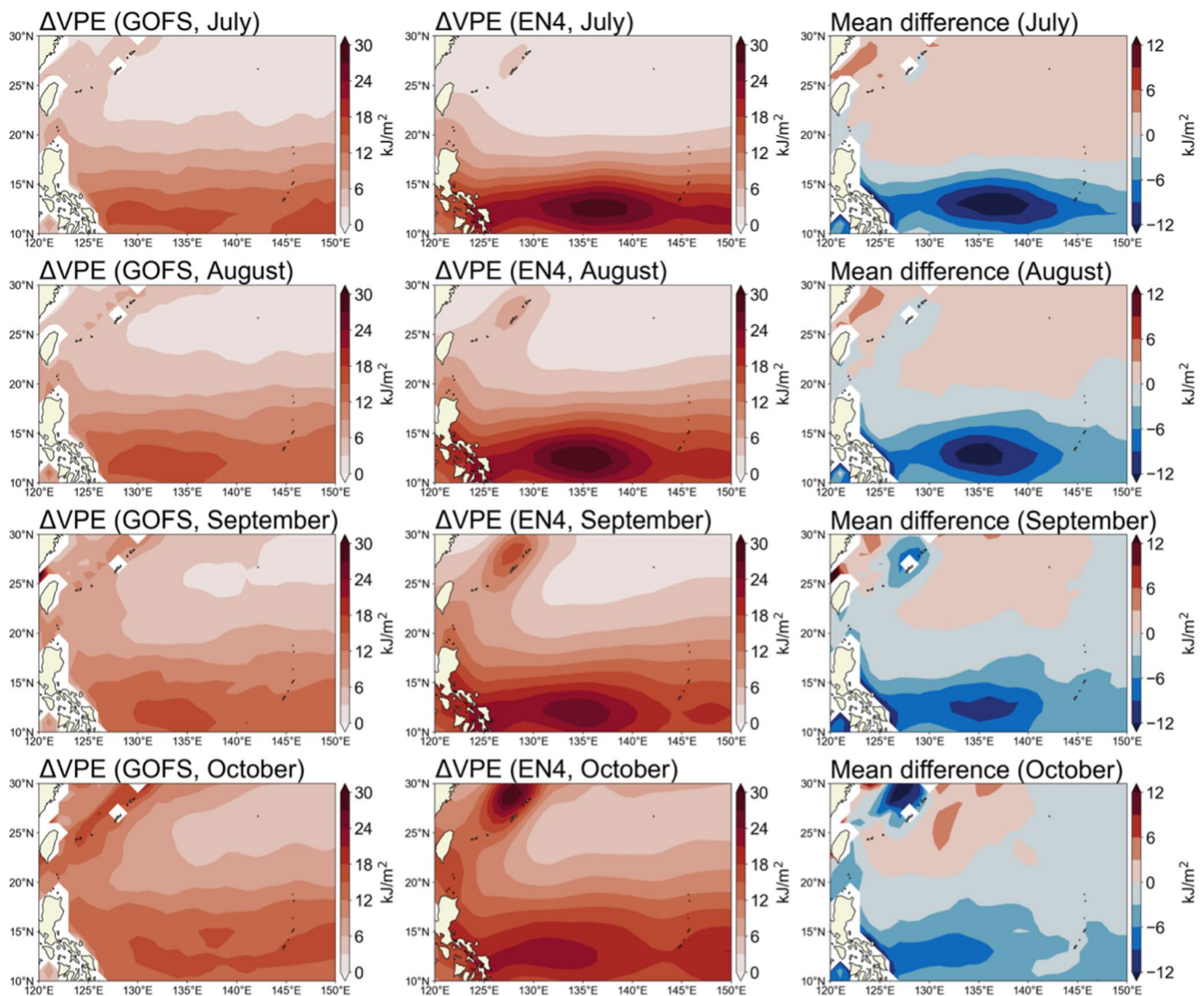


Fig. 8 As in Fig. 6, but for virtual potential energy change (ΔVPE) (kJ/m^2)

Table 2 Criteria* to classify surface wind energy (E_w), virtual potential energy change (ΔVPE), and TC translation speed (U)

E_w	Small	Large
	$E_{w20} \leq E_w \leq E_{w50}$	$E_{w50} \leq E_w \leq E_{w80}$
ΔVPE	$\Delta VPE_{20} \leq \Delta VPE \leq \Delta VPE_{50}$	$\Delta VPE_{50} \leq \Delta VPE \leq \Delta VPE_{80}$
U	$U_{30} \leq U \leq U_{70}$	

*The subscript m denotes the m th percentile of each parameter before classification

generally overestimates (underestimates) observed ΔVPE , associated with overestimated (underestimated) MLD and underestimated (overestimated) α (Figs. 6, 7 and 8). Although these results are based on monthly data and thus do not directly represent conditions during TC passages, they suggest that the difference between the classified large and small ΔVPE values in GOFS may be larger in observations. In other words, the difference in classified ΔVPE values in GOFS is unlikely to artificially enhance the corresponding difference in ΔSST .

3 Results

3.1 Relative importance of surface wind energy and upper-ocean stratification on surface cooling

First, we examine the impact of upper-ocean stratification on TC-induced sea surface cooling by comparing ΔSST between large and small ΔVPE groups with comparable E_w and U (Figs. 10a and b; Table S1). For selected strong

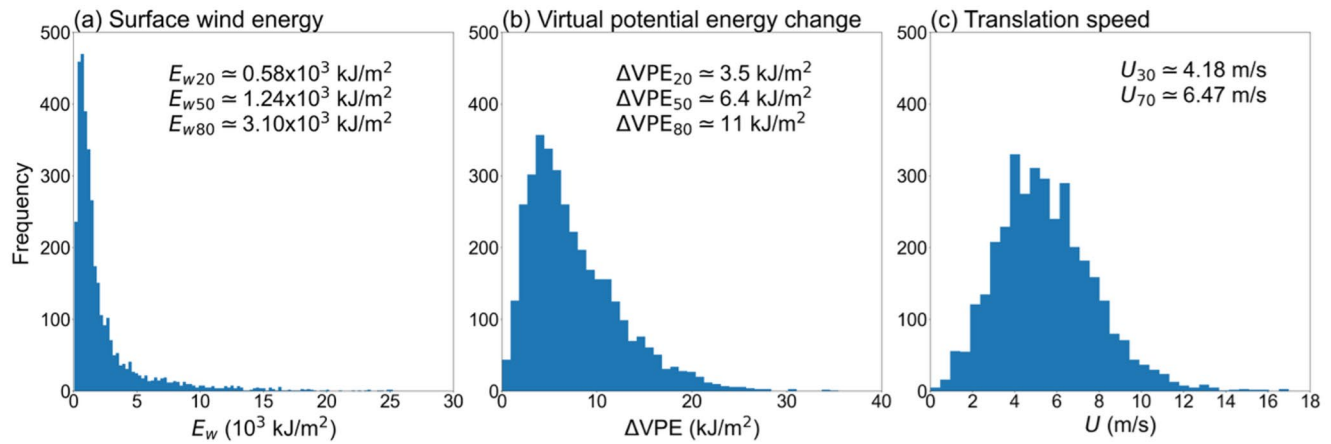


Fig. 9 Histograms of (a) E_w (kJ/m^2), (b) ΔVPE (kJ/m^2), and (c) TC translation speed (U) (m/s). The subscript m denotes the m th percentile of each parameter before classification

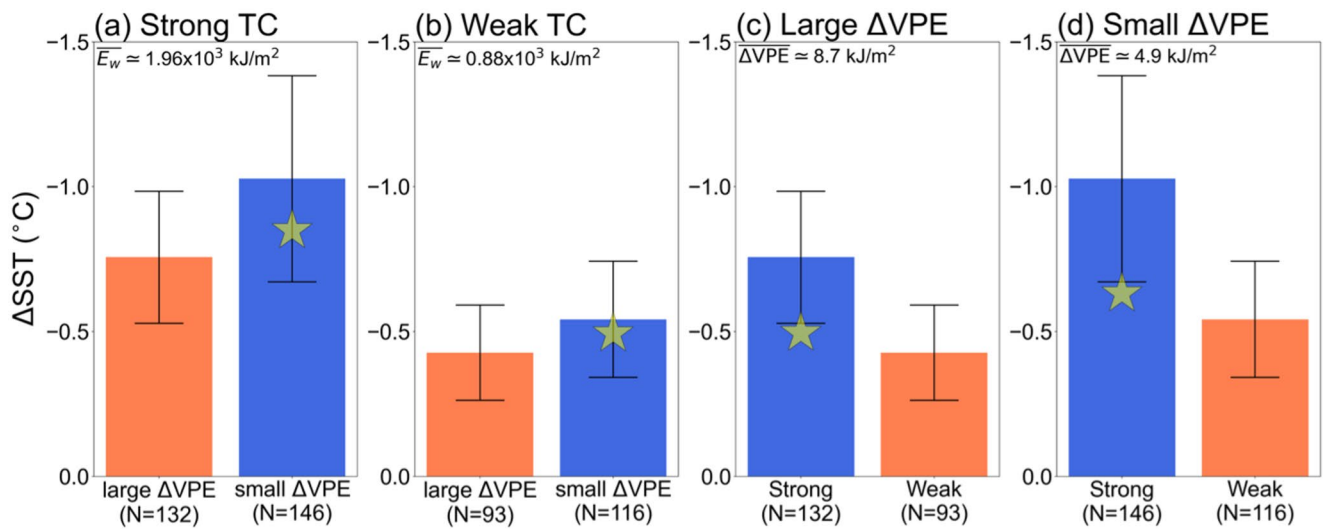


Fig. 10 Relationships between ΔVPE (kJ/m^2) and ΔSST ($^{\circ}\text{C}$) for (a) strong and (b) weak TCs, and those between E_w (kJ/m^2) and ΔSST for (c) large and (d) small ΔVPE . For each group, $\bar{E}_w$ or $\bar{\Delta \text{VPE}}$ is shown in the upper part, where the overbars represent the mean values. The orange and blue bars on panels a and b represent large and

small ΔVPE , whereas those on panels c and d represent weak and strong TCs, respectively. The number of samples N is indicated at the bottom. The error bars represent ± 1 standard deviation. Each yellow star represents the weakest cooling limit for ensuring the statistically significant difference in ΔSST between two subgroups at the 99% confidence interval by a two-tailed z-test

TCs (Fig. 10a), SST cools by 0.76°C when ΔVPE is large (8.6 kJ/m^2), whereas it cools by 1.03°C when ΔVPE is small (4.9 kJ/m^2). The corresponding mean E_w values are $2.02 \times 10^3 \text{ kJ/m}^2$ and $1.99 \times 10^3 \text{ kJ/m}^2$, and mean U values are 5.25 m/s and 5.41 m/s , respectively, which are not statistically different from each other. Here, the corresponding mean MLD values are 38.2 m and 30.2 m , and mean α values are -0.03 kg/m^4 and -0.04 kg/m^4 , respectively. Both differences are statistically significant.

These relationships are also evident for selected weak TCs (Fig. 10b). Specifically, SST cools by 0.43°C when ΔVPE is large (8.8 kJ/m^2), whereas it cools by 0.54°C when ΔVPE is small (4.7 kJ/m^2). The corresponding mean

MLD is 36.6 m and 29.9 m , and mean α is -0.03 kg/m^4 and -0.04 kg/m^4 , respectively. These differences are again statistically significant. In contrast, the corresponding mean E_w of $0.87 \times 10^3 \text{ kJ/m}^2$ and $0.86 \times 10^3 \text{ kJ/m}^2$, and mean U of 5.27 m/s and 5.22 m/s , respectively, are not statistically different from each other. Therefore, small ΔVPE , associated with a shallow MLD and a steep density gradient beneath the MLD, is more favorable for surface cooling. These results, based on reanalysis products, support the numerical findings of Vincent et al. (2012b).

We next investigate the impact of surface wind energy on surface cooling by comparing ΔSST between large and small E_w groups with comparable ΔVPE and U (Figs. 10c

and d; Table S1). For the large ΔVPE group (Fig. 10c), SST cools by 0.76°C when E_w is large ($2.02 \times 10^3 \text{ kJ/m}^2$), whereas it cools by 0.43°C when E_w is small ($0.87 \times 10^3 \text{ kJ/m}^2$), which are statistically different. Here, the corresponding mean ΔVPE values are 8.6 kJ/m^2 and 8.8 kJ/m^2 , and mean U values are 5.25 m/s and 5.27 m/s , respectively, which are not statistically different from each other. The corresponding mean MLDs are 38.2 m and 36.6 m , respectively, and the corresponding mean α values are both -0.03 kg/m^4 ; these variables are also not statistically different from each other.

Similar relationships are also seen for the small ΔVPE group (Fig. 10d). Specifically, SST cools by 1.03°C when E_w is large ($1.99 \times 10^3 \text{ kJ/m}^2$), whereas it cools by 0.54°C when E_w is small ($0.86 \times 10^3 \text{ kJ/m}^2$). These differences are statistically significant. The corresponding mean ΔVPE of 4.9 kJ/m^2 and 4.7 kJ/m^2 , U of 5.41 m/s and 5.22 m/s , MLD of 30.2 m and 29.9 m , and α of -0.04 kg/m^4 and -0.04 kg/m^4 , respectively, are comparable. Thus, surface cooling becomes stronger for stronger TCs under comparable upper-ocean stratification.

To further assess the relative importance of surface wind energy and upper-ocean stratification, we have linearly regressed ΔSST against E_w alone:

$$\text{regressed } \Delta \text{SST} = a \times E_w + c \quad (5)$$

and both E_w and ΔVPE :

$$\text{regressed } \Delta \text{SST} = a' \times E_w + b' \times \Delta VPE + c'. \quad (6)$$

Here, a , a' , and b' are partial regression coefficients, and c and c' are intercepts. Although E_w alone strongly correlates with ΔSST , the multiple regression improves the correlation coefficient from 0.78 to 0.85 , and reduces the RMSE from 0.74°C to 0.43°C (Fig. 11). The normalized partial regression coefficients for E_w and ΔVPE are -0.81 and 0.35 , respectively. These results indicate that although the difference or variance of TC-induced sea surface cooling is largely explained by surface wind energy, upper-ocean stratification is also important.

However, the multiple correlation coefficient of 0.85 indicates that a sizable fraction of the variance in ΔSST remains unexplained by the two predictors, E_w and ΔVPE . In addition to potential influences of TC translation speed, size, and other factors, nonlinear effects of E_w and ΔVPE (e.g., E_w^μ , ΔVPE^ν , and $E_w^\mu \Delta VPE^\nu$) may also affect the performance of the regression. Because it is difficult to isolate the individual contributions of these effects in a physically interpretable manner, we conduct a simpler analysis from the perspective of the independence between E_w and ΔVPE . If E_w and ΔVPE are completely uncorrelated or independent, their relative importance can be estimated using normalized partial regression coefficients or only partial correlation coefficients. However, the relatively low correlation coefficient between E_w and ΔVPE of 0.09 is not completely zero.

In this context, following the method of Lindeman et al. (1980) and assuming that ΔSST is explained solely by E_w and ΔVPE , we have calculated not only the partial correlation coefficient between ΔSST and E_w :

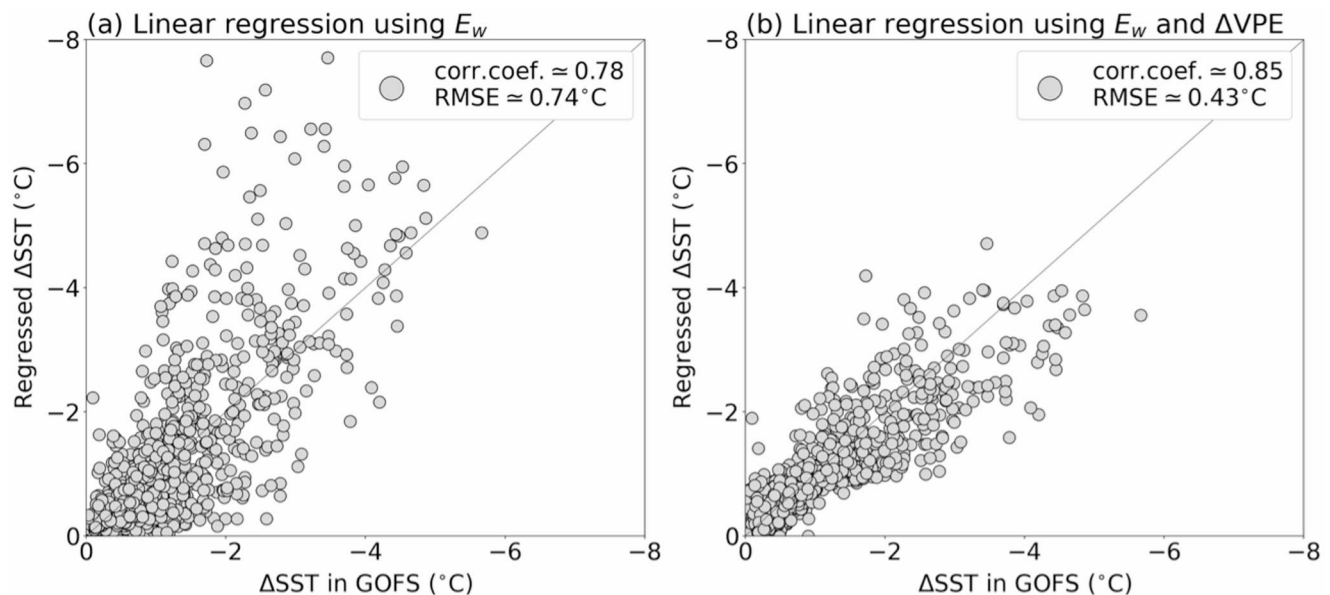


Fig. 11 Scatterplots comparing ΔSST obtained from GOFs with that estimated from linear regression analyses using (a) E_w alone and (b) both E_w and ΔVPE for TCs moving at translation speeds between

$U_{30} \approx 4.18 \text{ m/s}$ and $U_{70} \approx 6.47 \text{ m/s}$. Their correlation coefficient and RMSE between ΔSST obtained from GOFs and that estimated from each regression are shown in the upper right of each panel

$$= \frac{r_{\Delta \text{SST}, E_w} \cdot r_{\Delta \text{SST}, \Delta \text{VPE}} \times r_{E_w, \Delta \text{VPE}}}{\sqrt{(1 - r_{\Delta \text{SST}, \Delta \text{VPE}}^2)(1 - r_{E_w, \Delta \text{VPE}}^2)}}, \quad (7)$$

but also the *semi-partial correlation coefficient* between ΔSST and ΔVPE :

$$= \frac{r_{\Delta \text{SST}(\Delta \text{VPE} \cdot E_w)} \cdot r_{E_w, \Delta \text{VPE}}}{\sqrt{1 - r_{E_w, \Delta \text{VPE}}^2}}. \quad (8)$$

The left-hand side of Eq. (7) represents the correlation between ΔSST and E_w removing influences of ΔVPE on their variables, while the left-hand side of Eq. (8) represents the correlation between ΔSST and ΔVPE removing the relationship between E_w and ΔVPE . Each $r_{p,q}$ on the right-hand sides of these equations represents the correlation coefficient between p and q . Using the partial correlation coefficient $r_{(\Delta \text{SST}, E_w) \cdot \Delta \text{VPE}}$ and semi-partial correlation coefficient $r_{\Delta \text{SST}(\Delta \text{VPE} \cdot E_w)}$, we have estimated the percentage contributions of E_w and ΔVPE to the total variance in ΔSST as

$$P_{E_w} = \frac{r_{(\Delta \text{SST}, E_w) \cdot \Delta \text{VPE}}^2}{r_{(\Delta \text{SST}, E_w) \cdot \Delta \text{VPE}}^2 + r_{\Delta \text{SST}(\Delta \text{VPE} \cdot E_w)}^2} \times 100 \cong \frac{(-0.84)^2}{(-0.84)^2 + 0.35^2} \times 100 \cong 85 \text{ (\%)} \quad (9)$$

and

$$P_{\Delta \text{VPE}} = \frac{r_{\Delta \text{SST}(\Delta \text{VPE} \cdot E_w)}^2}{r_{(\Delta \text{SST}, E_w) \cdot \Delta \text{VPE}}^2 + r_{\Delta \text{SST}(\Delta \text{VPE} \cdot E_w)}^2} \times 100 \cong \frac{0.35^2}{(-0.84)^2 + 0.35^2} \times 100 \cong 15 \text{ (\%)}, \quad (10)$$

respectively.

Here, we note that the percentage contributions of E_w and ΔVPE are changed when using the partial correlation coefficient between ΔSST and ΔVPE , $r_{(\Delta \text{SST}, \Delta \text{VPE}) \cdot E_w}$, and the semi-partial correlation coefficient, $r_{\Delta \text{SST}(E_w \cdot \Delta \text{VPE})}$, due to the non-zero correlation between E_w and ΔVPE . Specifically, the percentage contributions of E_w and ΔVPE can also be estimated as

$$P'_{E_w} = \frac{r_{\Delta \text{SST}(E_w \cdot \Delta \text{VPE})}^2}{r_{\Delta \text{SST}(E_w \cdot \Delta \text{VPE})}^2 + r_{(\Delta \text{SST}, \Delta \text{VPE}) \cdot E_w}^2} \times 100 \cong \frac{(-0.81)^2}{(-0.81)^2 + 0.55^2} \times 100 \cong 68 \text{ (\%)} \quad (11)$$

and

$$P'_{\Delta \text{VPE}} = \frac{r_{(\Delta \text{SST}, \Delta \text{VPE}) \cdot E_w}^2}{r_{\Delta \text{SST}(E_w \cdot \Delta \text{VPE})}^2 + r_{(\Delta \text{SST}, \Delta \text{VPE}) \cdot E_w}^2} \times 100 \cong \frac{0.55^2}{(-0.81)^2 + 0.55^2} \times 100 \cong 32 \text{ (\%)}, \quad (12)$$

respectively.

As discussed above, since the percentage contributions of E_w and ΔVPE depend on their correlation, their relative importances on the variance in ΔSST cannot be simply determined from a single regression. In this regard, Lindeman et al. (1980) proposed estimating relative importance by averaging the percentage contributions over all possible permutations of predictors. Applying this method in this study, the percentage contributions of E_w and ΔVPE are estimated to be:

$$\frac{P_{E_w} + P'_{E_w}}{2!} \cong 77\% \quad (13)$$

and

$$\frac{P_{\Delta \text{VPE}} + P'_{\Delta \text{VPE}}}{2!} \cong 23\%, \quad (14)$$

respectively.

Based on the assumption that ΔSST is explained solely by E_w and ΔVPE , this methodology ultimately enables us to conclude that the percentage contributions of E_w and ΔVPE to the variance in ΔSST are 77% and 23%, respectively. These results indicate that the upper-ocean stratification plays a secondary but still substantial role in modulating TC-induced surface cooling.

3.2 Comparison across different upper-ocean measures

To quantify the impact of upper-ocean stratification on ΔSST , various upper-ocean measures have been used in previous studies. These include the density difference between the surface and the 100 m depth ($\rho_{\text{surface}} - \rho_{100 \text{ m}}$) (e.g., Looney and Foltz 2025) and vertically averaged potential temperature in the upper 100 m (T100) (e.g., Price 2009). We evaluate the performance of ΔVPE relative to other commonly used measures, through linear regressions of ΔSST against E_w and each metric (Table 3). The results show that ΔVPE exhibits the highest performance: the multiple regression using both E_w and ΔVPE reduces the RMSE by 41.2% compared with the simple regression using only E_w . Although the difference in ΔSST is significantly associated with that in MLD and α , the improvement of RMSEs

Table 3 Comparison across different upper-ocean measures*: Δ VPE, mixed layer depth (MLD), potential density gradient beneath MLD (α), potential density difference between the surface and the depth of 100 m ($\rho_{\text{surface}} - \rho_{100\text{ m}}$), and vertically averaged potential temperature in the upper 100 m (T100)

Measure	Correlation coefficient	RMSE ($^{\circ}\text{C}$)	Improvement of RMSE (%)
Δ VPE	0.276 (0.853)	1.16 (0.434)	41.2
MLD	0.109 (0.851)	0.923 (0.436)	40.9
α	0.253 (0.806)	0.835 (0.493)	33.3
$\rho_{\text{surface}} - \rho_{100\text{ m}}$	0.233 (0.828)	0.816 (0.467)	36.7
T100	0.190 (0.806)	0.844 (0.493)	33.3

*The correlation coefficient outside (in) each parenthesis is calculated between Δ SST and each measure obtained from GOFS (between Δ SST from GOFS and regressed Δ SST estimated from each multiple linear regression analysis using both the corresponding measure and E_w). The RMSE outside (in) each parenthesis is calculated between Δ SST from GOFS and regressed Δ SST estimated from each linear regression analysis using only the corresponding measure (between Δ SST from GOFS and regressed Δ SST estimated from each multiple linear regression analysis using both the corresponding measure and E_w). The improvement of RMSE (%) represents the percentage reduction in RMSE relative to the simple regression using only E_w

by including them is 40.9% and 33.3%, respectively, both of which are smaller than that achieved using Δ VPE. In addition, although $\rho_{\text{surface}} - \rho_{100\text{ m}}$ accounts for the impact of density gradients, the associated RMSE reduction is 36.7%, again lower than that using Δ VPE. This is probably because this metric omits the effect of mixed-layer thickness. Since oceanic density is nearly homogeneous from the surface to MLD, $\rho_{\text{surface}} - \rho_{100\text{ m}}$ primarily represents the density gradient beneath MLD. Furthermore, T100, a purely thermal measure excluding the effect of density stratification, yields an even smaller RMSE reduction of 33.3%. Therefore, Δ VPE appears to be the most effective measure for quantifying the impact of upper-ocean stratification on TC-induced sea surface cooling.

4 Summary and discussions

Using high-resolution reanalysis products and observed TC passages in the western North Pacific during July–October from 1994 to 2015, this study has investigated the relative importance of TC intensity and upper-ocean stratification in controlling sea surface cooling within 24 h before and after TC passages, with a focus on vertical mixing, a dominant process for the cooling. Slower-moving and larger TCs may also induce stronger cooling owing to longer forcing times (Liu et al. 2023; Looney and Foltz 2025). However, the incorporation of these effects would reduce the sample size to a point where their statistical significance cannot be verified. To ensure a sufficient number of cases for explicit

quantifications of TC intensity and upper-ocean stratification, we have retained TCs moving at typical speeds between 4.18 m/s and 6.47 m/s, without classifying TC size. TC intensity is defined by temporally integrated surface wind energy, whose integrand is proportional to the cube of surface wind speed. To characterize upper-ocean stratification, we have introduced Δ VPE, defined as the virtual potential energy change assuming a virtual surface cooling of -1°C . This measure incorporates the effects of both MLD and the density gradient beneath it. To account for the asymmetric distribution of strong surface cooling around the TC tracks, each variable was spatially averaged within a $2^{\circ}\times 2^{\circ}$ box centered on each grid point passed by a TC center.

The cooling amplitude is found to be larger for stronger TCs and for smaller Δ VPE, which is associated with a shallower MLD and a steeper density gradient beneath it. When investigating the influence of surface wind energy (upper-ocean stratification), the corresponding upper-ocean stratification (surface wind energy) values are made comparable between the classified subgroups. Specifically, for strong TCs with a mean surface wind energy of $1.96 \times 10^3 \text{ kJ/m}^2$, SST cools by 0.76°C when Δ VPE is large (8.6 kJ/m^2), whereas it cools by 1.03°C when Δ VPE is small (4.9 kJ/m^2). The corresponding mean MLDs are 38.2 m and 30.2 m, and the mean density gradients beneath the MLD are -0.03 kg/m^4 and -0.04 kg/m^4 , respectively. Conversely, for the small Δ VPE group with a mean value of 4.9 kJ/m^2 , SST cools by 1.03°C under large surface wind energy ($1.99 \times 10^3 \text{ kJ/m}^2$), whereas it cools by 0.54°C under small surface wind energy ($0.86 \times 10^3 \text{ kJ/m}^2$). All of these differences are statistically significant at the 99% confidence level.

Furthermore, linear regression analyses indicate that although E_w alone strongly correlates with Δ SST, the multiple regression using both E_w and Δ VPE improves the correlation coefficient from 0.78 to 0.85, and reduces the RMSE by 41.2% from 0.74°C to 0.43°C . Based on the assumption that Δ SST is explained solely by E_w and Δ VPE, using both partial and semi-partial correlation coefficients, the percentage contributions of E_w and Δ VPE to the variance in Δ SST are estimated to be approximately 77% and 23%, respectively. Therefore, although a large fraction of TC-induced sea surface cooling is explained by surface wind energy, the cooling is also controlled by upper-ocean stratification. Most previous studies used relatively low-resolution products, even though TC development is strongly influenced by cooling near the TC center on short timescales. In this context, using high-resolution reanalysis products and a classification method combined with regression analyses, this study quantifies SST variations under TCs and the potential impacts of both TC intensity and upper-ocean stratification on the cooling.

Although this study has focused on SST responses to TC forcing through upper-ocean stratification, the surface cooling is also recognized to provide a negative feedback to TC intensity. In this regard, Cione and Uhlhorn (2003) used observational data obtained from airborne XBTs together with fixed and drifting buoys, and statistically analyzed changes in SST and maximum surface wind speed near the eyewall for 23 hurricanes. They found that, for cases with small (large) cooling of 0.3°C (0.9°C), the maximum wind speed increased by 14.5 m/s (2.3 m/s), indicating that a surface cooling of 1°C near the TC center may suppress TC intensity by approximately 20 m/s. In the tropics, atmospheric convection deepens dramatically when SST exceeds about 27.5°C (Graham and Barnett 1987). Since the mean SST 24 h before TC passage is approximately 29°C across all classified TC events (not shown), TC development may be sensitive to even slight surface cooling through its modulation of moisture supply from the ocean.

Our findings are expected to offer valuable insights for practical TC intensity forecasts. For instance, in current forecasts by the Japan Meteorological Agency (JMA), annual mean RMSEs in maximum wind speeds are 5–10 m/s (Ono et al. 2019). Given that a surface cooling of 1°C near a TC center may suppress the TC intensity by 20 m/s, accurately predicting cooling influenced by upper-ocean stratification to within 0.25–0.5°C could potentially lead to substantial reductions in forecast errors. We have found that the mean SST change is 0.76 °C (1.03 °C) when Δ VPE is 8.6 kJ/m² (4.9 kJ/m²) for strong TCs, indicating that a 43% reduction in Δ VPE corresponds to an additional cooling of approximately 0.27°C. This effect of upper-ocean stratification on surface cooling may therefore contribute to improvements in current TC intensity forecasts.

Furthermore, through multiple regression analyses, we have verified the efficacy of Δ VPE relative to other commonly used upper-ocean measures, including MLD, the potential density gradient beneath it, potential density difference between the surface and 100 m, and vertically averaged potential temperature in the upper 100 m. Despite the simplicity of the regression framework, Δ VPE consistently outperforms these alternative measures in estimating the impact of upper-ocean stratification on TC-induced sea surface cooling. In addition to integrating the effects of both MLD and the density gradient beneath it, including contributions of thermal and haline stratification, Δ VPE is defined by virtual changes in SST and upper-ocean potential energy. Since TC-induced vertical mixing converts mechanical energy supplied by surface winds into an increase in upper-ocean potential energy (Liu et al. 2008), Δ VPE is more directly linked to the cooling process than metrics based solely on the thermal structure before a TC passage.

Table 4 As in Table 3, but for comparisons among Δ VPE, the ocean heat content (OHC) in the upper 100 m, and the tropical cyclone heat potential (TCHP)

Measure	Correlation coefficient	RMSE (°C)	Improvement of RMSE (%)
Δ VPE	0.276 (0.853)	1.16 (0.434)	41.2
OHC	0.190 (0.806)	0.844 (0.493)	33.3
TCHP	0.165 (0.790)	1.00 (0.510)	31.0

In light of this study, Δ VPE would be an additional predictor to enhance statistical and dynamical guidance models, such as the Typhoon Intensity Forecasting Scheme (TIFS) developed by JMA and operational since 2017 (Shimada 2024). This scheme employs multiple linear regressions based on atmospheric and oceanic predictors, achieving accuracy comparable to numerical models at relatively low computational cost. However, the effects of upper-ocean stratification in inhibiting TC development are represented only by the ocean heat content (OHC). Although OHC and tropical cyclone heat potential (TCHP) (Leipper and Volgenau 1972) are widely-used energy-based metrics for TC intensity forecasting, these measures are largely determined by thermal stratification, which is a function of the product of density and temperature:

$$OHC = \int_{100\text{ m}}^{0\text{ m}} \rho_i(T_i, S_i) C_p T_i dz \quad (15)$$

$$TCHP = \int_{D26}^{0\text{ m}} \rho_i(T_i, S_i) C_p (T_i - 26^\circ\text{C}) dz, \quad (16)$$

where ρ_i , T_i , S_i , and dz are defined as in Eq. (3), C_p is the specific heat of the seawater at constant pressure, and D26 is the depth of 26 °C isotherm. Following the standard definition, TCHP is non-zero only where $T_i \geq 26^\circ\text{C}$. In fact, multiple regressions using OHC or TCHP together with E_w exhibit lower performance than those using Δ VPE together with E_w (Tables 3 and 4). Specifically, the multiple regression using E_w and OHC (TCHP) reduces the RMSE by 33.3% (31.0%), compared with the simple regression using only E_w , whereas the corresponding reduction reaches 41.2% when Δ VPE is used. These results suggest that incorporating both mixed-layer thickness and the density gradient beneath it provides a more effective representation of upper-ocean stratification than thermal measures alone. Given that Δ VPE exhibits higher performance than OHC and TCHP in estimating TC-induced surface cooling, and that TC development is strongly affected by SST cooling beneath the TC (Emanuel 1999b; Cione and Uhlhorn

2003), our results obtained using Δ VPE provide a basis for improving the representation of ocean feedbacks in real-time TC intensity forecasts.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10872-026-00810-1>.

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Author contributions All authors contributed to the conceptualization. R. M. performed data analyses and prepared figures. R. M. wrote the first draft of the manuscript and T. T. contributed to the writing. All authors read and approved the final manuscript.

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Data availability TC positions were obtained from the best track data provided by the Regional Specialized Meteorological Center in Tokyo (<https://www.jma.go.jp/jma/eng/jma-center/rsmc-hp-pu-b-eg/besttrack.html>). To calculate TC intensity, surface wind speeds during 1994–2010 and 2011–2015 were obtained from the National Centers for Environmental Prediction (NCEP) Climate Forecast System Reanalysis (<https://www.hycom.org/dataserver/ncep-cfsr>) and the NCEP Climate Forecast System Version 2 (<https://www.hycom.org/dataserver/ncep-cfsv2>), respectively. To investigate upper-ocean stratification, temperature and salinity were obtained from the Global Ocean Forecasting System (GOFS) 3.1 (<https://www.hycom.org/dataserver/gofs-3pt1/reanalysis>). To validate the reanalysis datasets, observed wind speeds and upper-ocean responses associated with TC passages were obtained from the Kuroshio Extension Observatory (KEO) buoy (<https://www.pmel.noaa.gov/ocs/data/disdell/>). To further evaluate GOFS over a broad region passed by TCs, satellite-based observed sea surface temperatures were also obtained from the reprocessed Operational Sea Surface Temperature and Ice Analysis (OSTIA) system (https://data.marine.copernicus.eu/product/SST_GLO_SST_L4_REP_OBSERVATIONS_010_011/services). To investigate monthly mean upper-ocean structures, GOFS was also compared with the objective analyses from the EN 4.2.2 c14 (<https://www.metoffice.gov.uk/hadobs/en4/download-en4-2-2.html>).

Declarations

Conflict of interest The authors declare they have no competing interests that are relevant to the content of this article.

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